Juliana Oyola-Pabon

San Bruno, CA | julianaaoyola@gmail.com | LinkedIn | GitHub | Portfolio

Core Focus

Software Engineer specialized in frontend architectures and human-centered systems. Expertly maps computer science principles and cognitive psychology models onto high-scale data infrastructure, complex interface states, and web accessibility frameworks.

Skills & Tools

Frontend & Design: UI/UX Design, Interaction States, WCAG Accessibility, Figma, React, Next.js, TypeScript, Tailwind, Framer Motion
Backend & Cloud: Java, Spring Boot, Python, Node.js, REST APIs, Microservices, Systems Automation, Distributed Pipelines
Infrastructure: AWS (CDK, Lambda, API Gateway, DynamoDB, S3, CloudWatch), SQL (PostgreSQL, MySQL), Docker, CI/CD, Git

Professional Experience

Amazon
Seattle, WA

SOFTWARE ENGINEER | PRIME VIDEO
May 2024 – September 2025

- Centralized disparate internal metadata operations hosting millions of assets into a unified Java publishing framework, completely eliminating operational dependencies on siloed legacy tools.
- Engineered real-time ingestion pipelines and validation states for corrupt incoming datasets, preserving structural data schema continuity and preventing cascading failures down the publishing stack.
- Implemented repeatable web service patterns across AWS infrastructure utilizing AWS CDK, configuring explicit IAM policy matrices and CloudWatch triggers to optimize real-time incident resolution.
- Governed cross-functional design review loops between software groups and core systems operators, authoring systemic architectural documentation to reduce runtime engineer onboarding friction.

Amazon
Seattle, WA

SOFTWARE ENGINEER INTERN | PRIME VIDEO
May 2023 – August 2023

- Designed an automated backend normalization pipeline via Java and AWS CDK that scrubbed bad data at the application boundary, reclaiming roughly six hours of manual operational debugging per incident.

Amazon
Seattle, WA

SOFTWARE DEVELOPMENT ENGINEER INTERN | AMAZON CARE
May 2022 – August 2022

- Maintained critical transactional microservices handling user scheduling and asset distribution for a high-security digital healthcare application servicing over 25,000 monthly active users.
- Replaced legacy chronological execution paradigms with proximity-based matching routing systems, securing a 2× surge in platform engagement while slashing application abandonment by 30%.

Green Bank Observatory
Green Bank, WV

WEB DEVELOPER INTERN
January 2020 – May 2021

- Normalised backend data layers processing thousands of active telescope operations daily, boosting SQL index query execution efficiency by 30% while shrinking storage footprints by 20%.

Selected Product Work

MacWipe
PRODUCT DESIGN / SWIFT / APPKIT — View Project →

Built a native macOS storage engine around user safety boundaries. Decoupled diagnostic scanning from active execution, using explicit exclusion variables and confirmation dialog layers to prevent accidental file corruption.

GoogleCleaner
UX DESIGN / FRONTEND / JAVASCRIPT / DRIVE API — View Project →

Designed a bulk management tool built on persistent selection tracking. Maintained user state architecture cleanly across deep data sorting modifications, integrating background state workers to handle API request thresholds.

FigmaCleaner
PLUGIN DESIGN / TYPESCRIPT / FIGMA API — View Project →

Authored an internal workspace optimization layout tool. Consolidated manual layer nested operations into macro shortcuts.

Education

Florida State University
Tallahassee, FL

B.S. COMPUTER SCIENCE
Graduated May 2024

Santa Fe College
Gainesville, FL

A.A. PSYCHOLOGY (HONORS)
Transferred August 2020